Sr.No. 8373

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 202 / Course Section: 1	Seat No: 2022600011

1_61_469_986_1_1505_48243

Name of the candidate: Ansh Purnesh Gandhi

Candidate's UID number: 2 0 2 2 6 0 0 0 1 1

Examination class room number: 2 0 2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: B.E. BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE(AIML)

DAY AND DATE: Wednesday 08/10/2025

COURSE NAME: Deep Learning (DL)

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	
	Q1	
	(a)	Let the error function $f(w) = \frac{1}{2}\|\|Xw - y\|\|^2$

~~or~~ ~~Let~~ be $E(W) = \frac{1}{2} \|\| t_d - O_d \|\|^2$

$W_i \leftarrow W_i + \Delta W_i$ where W_i are weights

$\Delta W_i = -\eta \frac{\partial E(W_i)}{\partial w_i}$ & $\bar{O} = \bar{W} \cdot \bar{x}$.

$\therefore \frac{\partial E(W_i)}{\partial w_i} = \frac{1}{2} \frac{\partial \|\| t_d - O_d \|\|^2}{\partial w_i}$

$= \frac{1}{2} \times 2 \|\| t_d - O_d \|\| \times \frac{\partial \|\| t_d - O_d \|\|}{\partial w_i}$

$\frac{\partial E(W_i)}{\partial w_i} = \|\| t_d - O_d \|\| x_{id}$

$\therefore \Delta W_i = -\eta (t_d - O_d) x_{id}$

$\therefore$ The weight updation in gradient rule will take place as follows:-

$W_i \leftarrow W_i + -\eta (t_d - O_d) x_{id}$

$W_i \leftarrow W_i - \eta (t_d - O_d) x_{id}$

where η is the learning rate
t_d is the target variable
d is the number of data pairs.

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	Q1	
	(b)	Overfitting occurs

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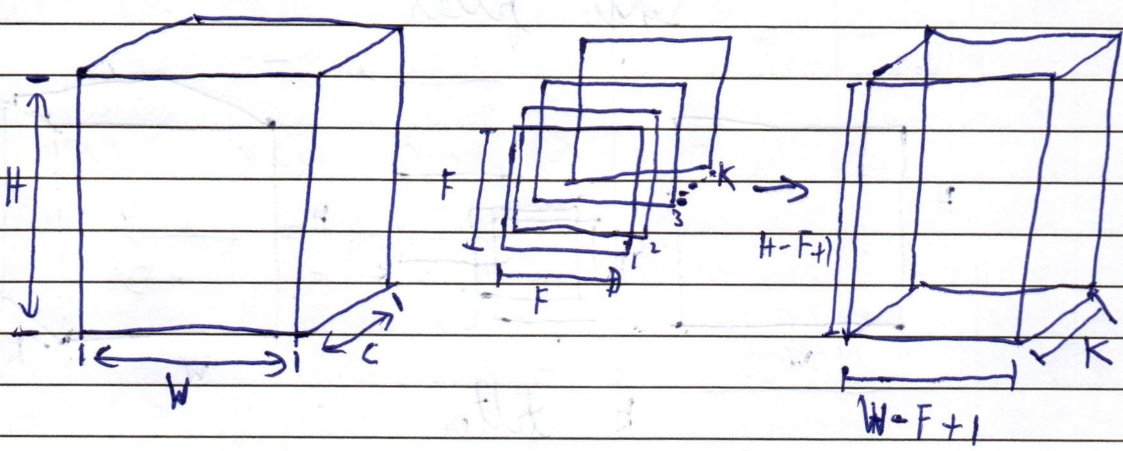
	Q2 (a)	~~For~~ a 2D convolution

For a depthwise separable convolution layer of input size $H \times W \times C$ and K filters of $F \times F$:-



The computational complexity for such operation = $H \times W \times F \times K \times C$

and the output dimensions = $(H-F+1) \times (W-F+1) \times K$

In the ~~input~~ filter number of weight will be equal to $F \times F$.

Number of bias per filter is 1

∴ parameters of 1 filter = $F \times F + 1$

Number of filters = K

∴ parameters = $K * (F \times F + 1)$

Number of values in input image = $H \times W \times C$

∴ Number of total parameters = $\cancel{H \times W \times C \times K}$ $H \times W \times C \times K * (F \times F + 1)$

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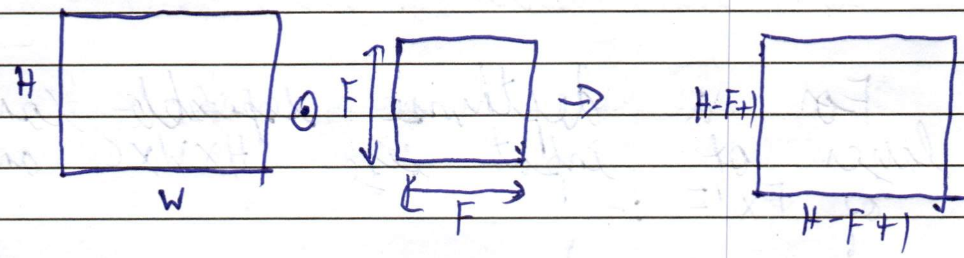
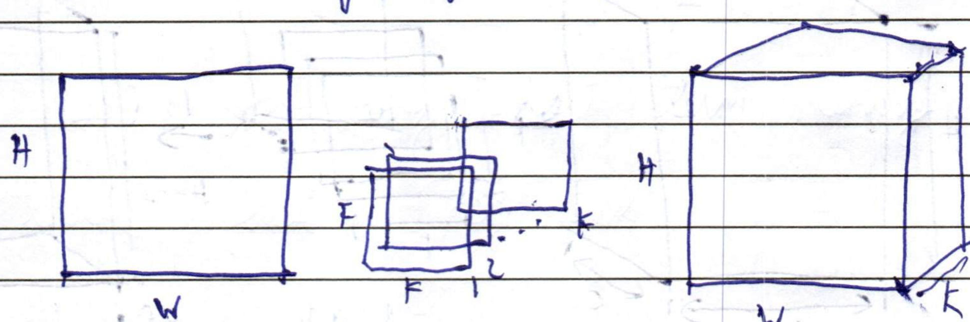
For a 2D-convolution layer



Single filter.



K filter

No. of parameters in filter $= F \times F$

~~if more than~~ with bias $= F \times F + 1$

If more than 1 filter $= K * (F \times F + 1)$

$\therefore$ Total number of parameters $= H \times W \times K \times (F \times F + 1)$

Depthwise separable convolutions are critical because they increase the overall accuracy of the output.

The pixels are more informative in the output than the 2D.

They also decrease underfitting problem.

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Q2

(b) Over complete encoders are the encoders which have the latent space equal to or greater than the input size or dimension.

Therefore if an input is provided to the over comple encoder ~~it will~~ without regularisation, it will tend to learn the identity function, it will copy paste the input space as it is to the latent space.

An example of this can be taken as we asked someone to give us a summary of a book and they gave us the book as it is as summary.

Thus to overcome this problem regularization is done is which a ~~loss~~ a penalty is added to the loss function to avoid such problems.

Q3(a) In RNN

$a_t = W_{hh} h_{t-1} + W_{xh} x + b_t$

$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x + b_t)$

$O_t = W_{hy} h_t + b_y$

$y_t = \text{softmax}(O_t)$.

LSTM or Long Short term memory have three gates. They are

- Forget Gate
- Input Gate
- Output Gate.

These gates helps the model to have long as well as the short memory unlike RNN.

The forget Gate decides what all values to be remembered from the past hidden state and what all to forget. It is made up of '0s' and '1s'. 0 means the value to be forgotten and 1 is remembered.

Thus this prevent vanishing gradients. Vanishing gradient occured because the weights W_{hh}, W_{xh} etc in RNN were fixed and if they were less (less than 1), the multiplication of gradient over a number of state would be almost equal to zero.

But in LSTM this problem is solved

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Let $z_t = [x_t, h_{t-1}]$

$\therefore f_t = \sigma(W_f z_t + b_f)$

$i_t = \sigma(W_i z_t + b_i)$

$O_t = \sigma(W_o z_t + b_o)$

$\bar{c}_t = \tanh(W_c z_t + b_c)$

$c = f_t \odot c_{t-1} + \cancel{i_t \odot \bar{c}_t}\ o_t \odot \bar{c}_t$

Thus it can be seen that forget gate gives an answer between 0 & 1 and also ensures which value to be remembered and which to forget from the previous candidate cell.

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Q3 (b)

Let $z_t = [x_t, h_{t-1}]$ ... (i)

$\therefore u_t = \sigma(W_u z_t + b_u)$ ... (ii)

$r_t = \sigma(W_r z_t + b_r)$ ... (iii)

$\bar{c}_t = \sigma(W_c * r_t z_t + b_c)$ ... (iv)

$c_t = (1-u_t)^* c_{t-1} + \bar{c}_t u_t$

Here u_t ~~is~~ refers update gate
r_t refers to reset gate
~~&~~ $\bar{c}_t$ refers to candidate cell.

As it can be seen from the equation (ii) & (iv)

the update gate acts as a convex combination of previous hidden state and candidate hidden state

Equation (ii) ~~takes~~ decides information from previous hidden state

(iv) ~~equation allows either previous candidate~~ ~~equati~~ information

equation (iv) allows only either of the two information to come. ~~That is~~ ~~current candidate~~ i.e. previous candidate ~~state~~ cell with 1- update gate or update gate and $\bar{c}_t$.

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GRUs tend to have fewer ~~parametes~~ parameters because they have one lesser gate and they are less robust.

The update gate is a convex combination of previous hidden state and candidate hidden state, thus reducing parameters.

GRUs are used where efficiency and faster calculations are needed in case of small dataset.

They are efficient and faster.

For example stock market prediction of 20 days

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